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# Speed and Torque Control of an SRM: Mathematical and Simulation Modelling

By

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Submitted to

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# Chapter 1

# Introduction and Background

## 1.1 Introduction

This report serves as the technical continuation of Milestone 3, transitioning from the foundational literature review of Switched Reluctance Motor (SRM) technologies to the formal mathematical modeling and simulation phase. The primary objective is to validate the speed and torque control performance of a specific three-phase 6/4 SRM using an integrated MATLAB/Simulink environment.

The 6/4 Switched Reluctance Motor was selected for this research due to its magnet-free rotor construction, which offers significant advantages in thermal robustness and supply chain independence from rare-earth materials. This configuration provides an optimal balance between torque density and control simplicity for academic and industrial evaluation. This document details the derivation of the non-linear plant equations, the modular implementation of the Simulink subsystems, and the analysis of results under various loading conditions.

## 1.2 SRM Fundamentals and Comparative Analysis

### 1.2.1 Working Principle

The SRM operates on the principle of variable reluctance. Torque is produced by the tendency of the rotor poles to align with the energized stator poles to reach a state of minimum magnetic reluctance. A fundamental characteristic of the SRM is that torque production is independent of the direction of current flow; it depends solely on the rotor position and the rate of change of inductance ($dL/d\theta$). As the rotor moves from an unaligned to an aligned position, the inductance increases, and the energy converted into mechanical work generates motoring torque.

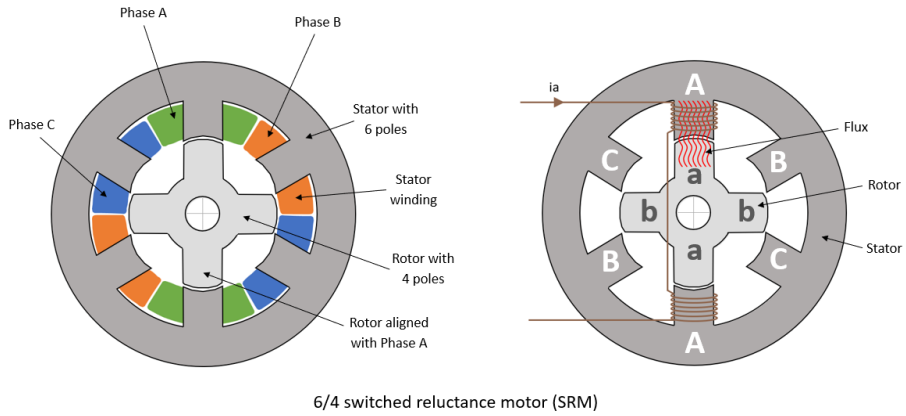



Figure 1.1: Cross-section and Flux Distribution of a 6/4 Switched Reluctance Motor. The left illustration shows the concentrated windings of the three phases, while the right depicts the flux path and rotor alignment in the low-reluctance position during Phase A excitation [1].

### 1.2.2 Comparative Analysis with Popular Motor Types

The SRM distinguishes itself from Induction Motors (IM) and Permanent Magnet Synchronous Motors (PMSM/BLDC) through several structural and operational differences:

- **Induction Motors (IM):** Unlike IMs, which require rotor windings or a squirrel cage that incur secondary copper losses, the SRM rotor consists purely of a stack of steel laminations. This eliminates rotor losses and simplifies the thermal management of the machine.
- **Synchronous Motors (PMSM/BLDC):** These machines depend on rare-earth magnets (e.g., NdFeB), which are vulnerable to thermal demagnetization and supply chain volatility. The SRM eliminates magnets entirely, ensuring reliability in high-temperature or hostile environments.
- **Winding and Flux:** IMs and PMSMs generally utilize distributed windings to produce sinusoidal flux. In contrast, the SRM uses concentrated windings on the stator poles, which simplifies the manufacturing process and enhances phase-to-phase fault tolerance.

### 1.2.3 SRM Topologies and Configurations

SRM configurations are defined by the ratio of stator poles ($N_s$) to rotor poles ($N_r$). The specific configuration dictates the number of phases and the resulting torque characteristics:

- **6/4 SRM (3-Phase):** This configuration utilizes 6 stator poles and 4 rotor poles. It is the focus of this study due to its high torque-to-volume ratio and straightforward power electronic requirements (3-leg asymmetric bridge).
- **8/6 SRM (4-Phase):** This topology provides smoother torque delivery and reduced ripple

compared to the 6/4 model but requires a more complex 4-leg inverter and additional sensors.

- **12/8 SRM (3-Phase):** A higher pole-count machine that shares the same 3-phase inverter as the 6/4 machine but offers reduced torque ripple at the expense of increased switching frequency and iron losses.

# Chapter 2

# Mathematical Modeling

## 2.1 Introduction

The Switched Reluctance Motor (SRM) is a doubly salient, singly excited machine that operates on the principle of minimum magnetic reluctance. This chapter derives the non-linear equations governing the electrical and mechanical states of a 3-phase 6/4 configuration. These derivations serve as the theoretical foundation for the Simulink blocks detailed in Chapter 3.

## 2.2 Electrical State Dynamics

The voltage balance for each of the three stator phases $k \in \{1, 2, 3\}$ is defined as:

$$v_k = R_s i_k + \frac{d\psi_k(\theta, i_k)}{dt} \tag{2.1}$$

In a 6/4 motor, the flux linkage $\psi_k$ is a non-linear function of both the current and the rotor position. By applying the chain rule, we decompose the flux derivative into transformer and rotational components:

$$v_k = R_s i_k + L(\theta)\frac{di_k}{dt} + i_k \omega \frac{dL(\theta)}{d\theta} \tag{2.2}$$

Physically, the term $i_k \omega \frac{dL}{d\theta}$ represents the **back-electromotive force (Back-EMF)**. The numerical integration of this derivative is handled by the discrete plant blocks explained in Section 3.2.

## 2.3 Inductance Profile and Motoring Region

The inductance $L(\theta)$ is periodic over the rotor pole pitch, which for a 6/4 machine is $90^\circ$ ($\pi/Nr$). It is modeled as a piecewise function varying between $L_{min}$ (unaligned) and $L_{max}$ (aligned). Motoring torque is strictly generated during the rising region where $dL/d\theta > 0$. The implementation of this profile using Look-Up Tables is detailed in Section **??**.

Figure 2.1: Theoretical Inductance Profile of a 6/4 SRM highlighting the motoring zones.

## 2.4 Torque and Motion Dynamics

The electromagnetic torque $T_{e,k}$ produced by a single phase is given by:

$$T_{e,k} = \frac{1}{2} i_k^2 \frac{dL_k(\theta)}{d\theta} \tag{2.3}$$

The mechanical state is governed by the second-order equation of motion:

$$J\frac{d\omega}{dt} = \sum_{k=1}^{3} T_{e,k} - T_L - B\omega \tag{2.4}$$

where $T_L$ is the external load torque. The modular realization of this mechanical system is described in Section 3.2.2.

# Chapter 3

# Simulation Model Implementation

The control system for the 6/4 SRM is characterized by its algorithmic implementation. While the mechanical speed loop uses standard discrete blocks, the complex electromagnetic logic, including torque distribution, reference inversion, and commutation is encapsulated within specialized **MATLAB Function blocks**. This modular approach ensures that the control logic is easily modifiable through the central initialization script. The full Simulink model is shown in 3.4

## 3.1 The Parameter Initialization Script

The simulation is driven by a specialized MATLAB script, `SRM_params.m`. This file acts as the configuration layer of the project, defining the physical constants of the 6/4 motor ($R_s, L_{max}, L_{min}, J, B$) and the control parameters ($K_p, K_i, \Delta i$). By centralizing these variables, the model supports rapid testing of different motor topologies and inertia settings without requiring architectural changes to the Simulink canvas.

## 3.2 Dynamic Plant Implementation

The motor plant is implemented using a modular subsystem approach to mirror the physical construction of the 6/4 machine.

### 3.2.1 Phase Subsystems and Electrical Dynamics

Each phase is modeled as an independent subsystem implementing the rearranged form of Equation 2.2.

- **Inductance Implementation:** As discussed in Section 1.3, the profile is realized via a 1-D Look-Up Table (LUT) mapping position $\theta$ to $L(\theta)$.
- **Back-EMF Calculation:** A product block calculates the rotational voltage term $i \cdot \omega \cdot \frac{dL}{d\theta}$.

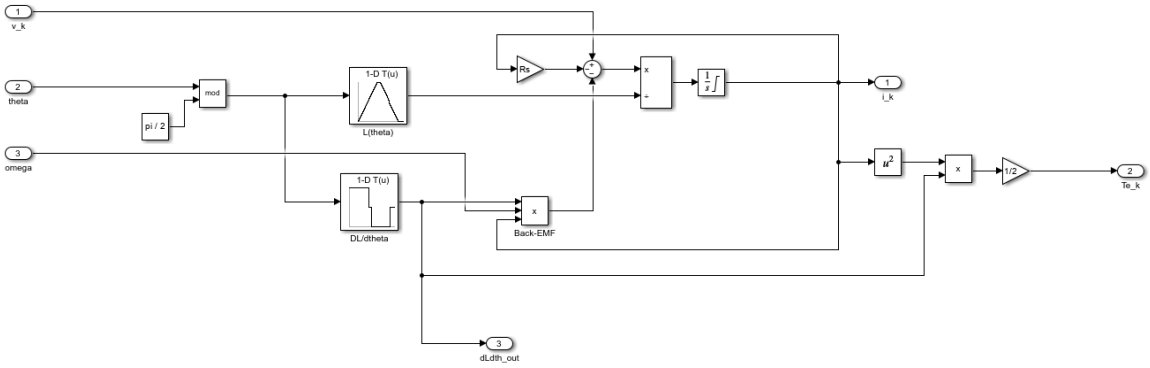



Figure 3.1: Simulink implementation of the 6/4 SRM Phase Dynamics subsystem. The model realizes the electrical state equation by integrating the derivative of current ($di/dt$), calculated as the net voltage (source minus resistive drop and back-EMF) divided by the instantaneous inductance $L(\theta)$ provided by the 1-D Look-Up Table. The lower branch implements the quadratic torque production logic: $T_e = \frac{1}{2} i^2 \frac{dL}{d\theta}$.

### 3.2.2 Mechanical Dynamics Subsystem

The mechanical block implements Equation 2.4. It aggregates the instantaneous torques from all three phases and subtracts the load torque $T_L$, which is modeled as a **Step block** to evaluate disturbance rejection at $t = 0.1s$. The resulting acceleration is integrated twice to obtain rotor velocity ($\omega$) and position ($\theta$).

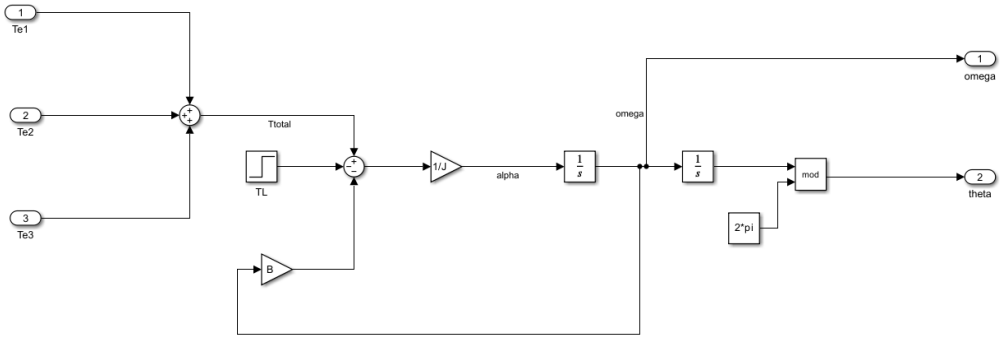



Figure 3.2: Simulink implementation of the Mechanical Dynamics subsystem. The model aggregates the three individual phase torques and subtracts the external load torque ($T_L$) and viscous friction ($B\omega$). The resulting net torque is processed through an inertial gain ($1/J$) and integrated to determine the rotor speed ($\omega$) and mechanical position ($\theta$).

## 3.3 Control Architecture Implementation

The control of the 6/4 SRM is implemented through a cascaded loop architecture, necessitated by the highly non-linear and coupled nature of the electromagnetic plant. The system is architected into two distinct layers to separate the mechanical and electrical time constants:

1. **Outer Speed Loop:** A low-bandwidth control loop that regulates the rotor velocity $\omega$. It generates the total required electromagnetic torque reference $T^*$.
2. **Inner Current Loops:** High-bandwidth loops designed to track the instantaneous current references $i_k^*$ for each phase. These loops must overcome the position-dependent back-EMF and inductive transients.

The interaction between these loops, the Torque Sharing Function (TSF), and the physical plant is illustrated in the block diagram in Figure 3.3.

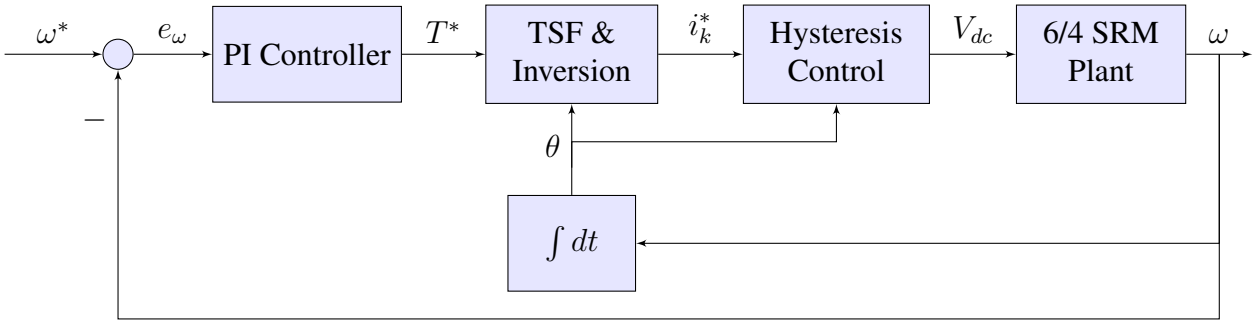



Figure 3.3: Cascaded control architecture for the 6/4 SRM featuring outer-loop speed regulation and inner-loop current control. The diagram illustrates the integration of speed and position feedback through a Torque Sharing Function (TSF) and hysteresis regulators to manage the non-linear motor dynamics.

### 3.3.1 Discrete Speed PI Controller

The outer control loop utilizes a *Discrete PI Controller* block to regulate the mechanical velocity. It processes the error between the reference speed $\omega^*$ and the measured feedback $\omega$ to generate a total electromagnetic torque reference $T^*$. All controller gains ($K_p, K_i$) and the target speed are extracted directly from the `SRM_params.m` workspace. This block serves as the system's primary regulator for rejecting load disturbances, which are also defined via variables in the parameter file.

### 3.3.2 MATLAB Function: TSF and Torque-to-Current Inversion

The bridge between total torque demand and individual phase excitation is implemented within a **MATLAB Function block**. This block handles the non-linear distribution of torque and its mathematical conversion to current references.

- **Programmatic TSF Selection:** The function evaluates the `TSF_type` variable assigned in the initialization script. It utilizes conditional logic to calculate individual torque references $T_k^*$ based on the selected profile:
  - **Linear TSF:** Distributes torque based on a linear ramp during the overlap interval, offering a simple computational path.
  - **Cubic TSF:** Employs a third-order polynomial to ensure that the derivatives of the torque references are zero at the commutation boundaries, providing a smoother transition and reducing torque ripple.
  - **Sinusoidal/Exponential TSF:** Additional profiles are available for testing specific ripple-mitigation trade-offs.
- **Non-linear Inversion:** To solve the non-linear relationship between torque and current, the function computes the current magnitude $i_k^* = \sqrt{2T_k^*/(dL/d\theta)}$. Performing this calculation programmatically ensures high numerical precision and avoids the complexity of manual math block wiring.

### 3.3.3 MATLAB Function: Phase Commutation Logic

Synchronization of the stator phases is handled by a dedicated **MATLAB Function block** acting as an electronic commutator. This logic ensures the motor operates strictly within the motoring region.

- **Position Wrapping:** The function utilizes the `mod(theta, pi/2)` operator to resolve the continuous mechanical rotor position into the specific $90^\circ$ pole pitch of the 6/4 SRM.
- **Gating Logic:** The turn-on ($\theta_{on}$) and turn-off ($\theta_{off}$) angles are retrieved from the parameter script. The function evaluates the wrapped position against these thresholds to generate binary gating signals. This prevents the generation of negative (braking) torque by ensuring energy is only injected when the inductance gradient is positive.

### 3.3.4 MATLAB Function: Hysteresis Current Regulation

The innermost control layer consists of independent **MATLAB Function blocks** that simulate the switching behavior of the power converter. These functions implement the following hysteresis logic:

$$V_{out} = \begin{cases} +V_{dc} & \text{if } (i_k^* - i_k) > \Delta i \\ -V_{dc} & \text{if } (i_k^* - i_k) < -\Delta i \end{cases} \tag{3.1}$$

Both the DC bus voltage ($V_{dc}$) and the hysteresis band ($\Delta i$) are extracted from the global parameters file. This decoupling allows for the rapid study of switching frequency impacts and current ripple simply by updating the workspace variables.

## 3.4 Solver and Simulation Settings

To ensure the stability of the high-frequency switching and the non-linear plant dynamics, the model utilizes a **Fixed-step ode4 (Runge-Kutta)** solver. The step size is globally defined at $10\mu s$. This high-resolution setting is mandatory to prevent numerical oscillations during the inductance transitions and to ensure the discrete PI controller updates fast enough to maintain speed regulation during load steps.

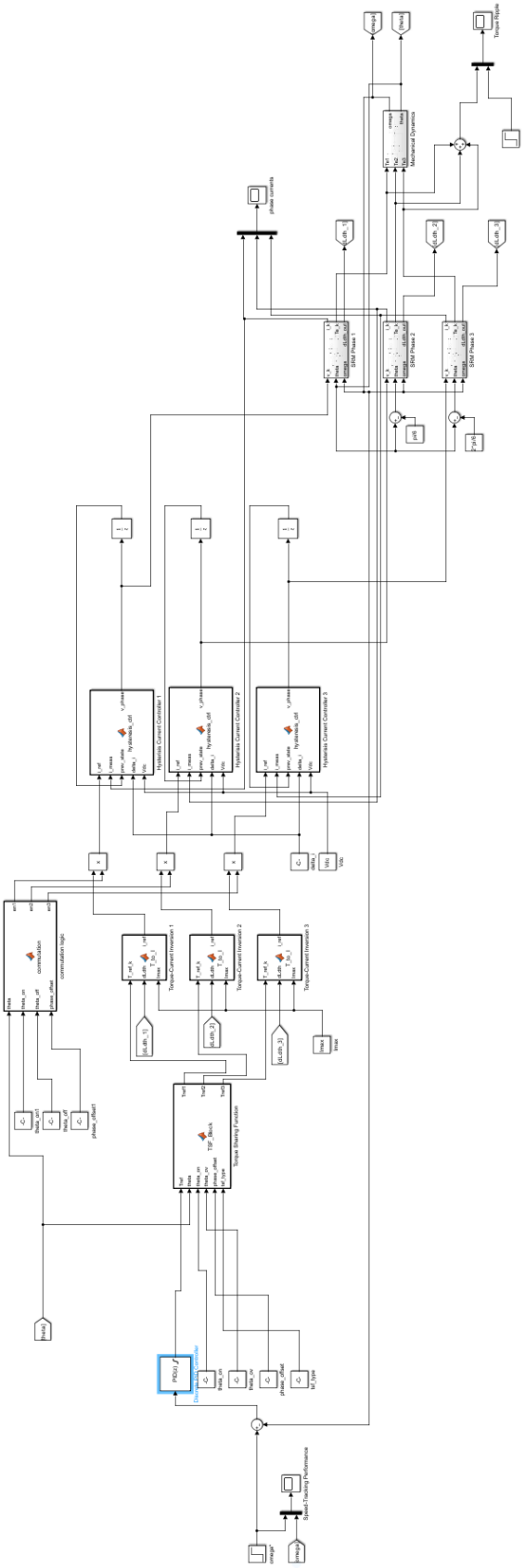


Figure 3.4: System-level architecture of the cascaded 6/4 SRM control platform, illustrating the integration of the discrete speed regulation loop with programmatic torque distribution and non-linear machine dynamics.

# Chapter 4

# Results and Discussion

This chapter details the simulation-based validation of the 6/4 SRM control platform. The analysis focuses on the implementation of the Cubic Torque Sharing Function (TSF 3) and evaluates its performance during high-inertia startup and under the influence of sudden load disturbances.

## 4.1 Simulation Environment and Parameters

The simulation was executed in a high-fidelity Simulink environment using a fixed-step **ode4 (Runge-Kutta)** solver. A fundamental step size of **10** $\mu$**s** was maintained to accurately resolve the high-frequency switching transients of the hysteresis current loops and the programmatic transitions of the control logic. The physical and control parameters are summarized in Table 4.1.

Table 4.1: Motor and Control System Simulation Parameters

| **Parameter** | **Value** | **Parameter** | **Value** |
|---|---|---|---|
| DC Bus Voltage ($V_{dc}$) | 300 V | Hysteresis Band ($\Delta i$) | 0.5 A |
| Reference Speed ($\omega^*$) | 100 rad/s | Load Torque ($T_L$) | 1.5 N.m |
| Stator Resistance ($R_s$) | 2.0 $\Omega$ | Rotor Inertia ($J$) | 0.005 kg.m$^2$ |
| Viscous Friction ($B$) | 0.01 N.m.s/rad | Commutation Window | $90^\circ$ / Phase |

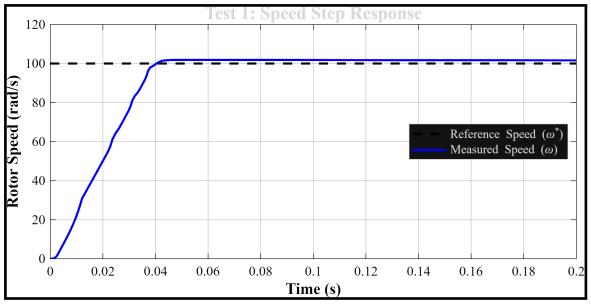
## 4.2 Interpretation of Graphical Results

To ensure a comprehensive understanding of the system's mechatronic health, the results are presented through three distinct plot types. Each provides specific insight into a different layer of the control hierarchy:

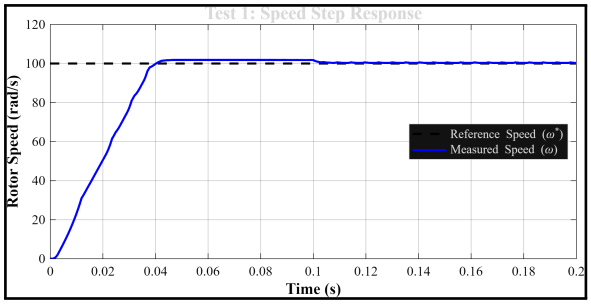
- **Rotor Speed Response ($\omega$):** Represents the macro-level performance. It tracks the ability of the outer PI loop to reach the target velocity and maintain it despite external torque application.
- **Phase Currents ($i_k$):** Represents the control effort. These plots illustrate the programmatic "S-curve" shaping of the current references. The smooth shoulders of the pulses indicate a controlled handover between stator poles.
- **Electromagnetic Torque ($T_e$):** Represents the output quality. This signal combines the torque from all active phases. The primary goal is to minimize the "dips" during commutation to ensure smooth mechanical rotation and low acoustic noise.

## 4.3 Performance Analysis of Cubic TSF (TSF 3)

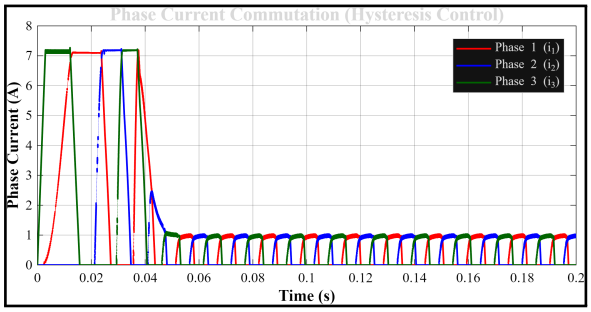
The following results demonstrate the system behavior under no-load and nominal load conditions. Figure 4.1 provides the consolidated diagnostic data for the Cubic TSF implementation.



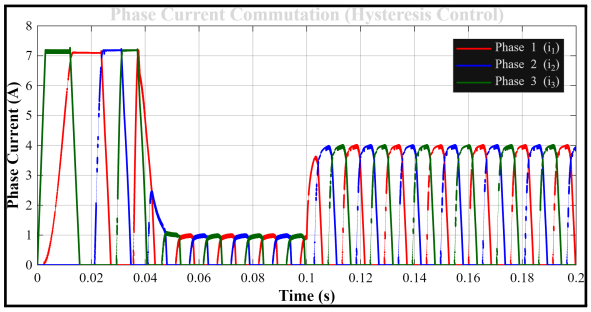

(a) Speed Response (No-Load)



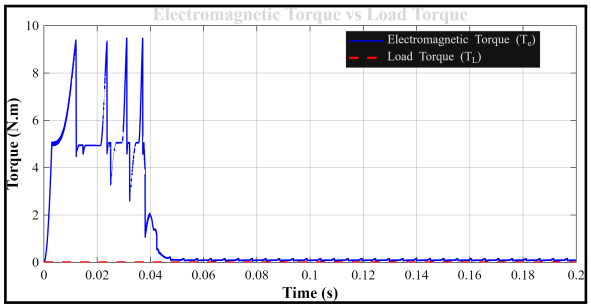

(b) Speed Response (1.5 N.m Load)



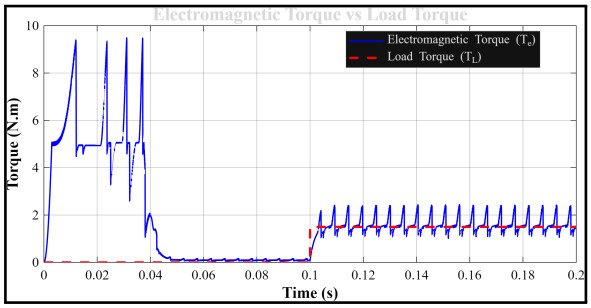

(c) Phase Currents (No-Load)




(d) Phase Currents (1.5 N.m Load)




(e) Torque Ripple (No-Load)




(f) Torque Ripple (1.5 N.m Load)

Figure 4.1: Performance characteristics of the 6/4 SRM utilizing Cubic TSF logic, illustrating the synchronization between mechanical velocity, programmatic phase excitation, and total electromagnetic torque under no-load and nominal load conditions.

## 4.4 Discussion

The simulation results provide a clear window into the dynamic behavior of the motor plant and the efficacy of the control architecture under different loading conditions.

### 4.4.1 Transient Startup and Controller Saturation

A prominent feature in the torque plots is the initial electromagnetic torque spike, reaching approximately 9 N.m during the startup interval ($0 < t < 0.04$ s). This phenomenon is a direct result of the PI controller saturating to minimize the large initial speed error. To reach the 100 rad/s setpoint as fast as possible, the speed loop commands maximum torque, causing the phase currents to hit their peak magnitudes. Once the target velocity is reached, the controller exits saturation, and the torque settles to the level required to overcome internal friction.

### 4.4.2 Load Disturbance and Speed Recovery

At t = 0.1 s, a nominal load of 1.5 N.m is applied to the motor shaft. This results in a momentary speed dip visible in Figure 4.1b. The integral term of the discrete PI controller identifies the steady-state error and compensates by increasing the total torque reference. This is evidenced by the increased magnitude of the phase current pulses in Figure 4.1d. The system demonstrates high robustness, restoring the reference speed within 0.02 s.

### 4.4.3 Commutation Logic and Hysteresis Noise

The interaction between the MATLAB-coded TSF and the hysteresis regulators defines the quality of the torque production:

- **Current Shaping:** The phase currents exhibit the smooth ”S-curve” shoulders characteristic of the Cubic TSF. This programmatic shaping ensures that the derivative of the torque reference is zero at the commutation boundaries, allowing for a mathematically smoother handover between phases.
- **Torque Dips:** Periodic dips are observed in the total electromagnetic torque during phase transitions. These are the result of the non-linear inductance gradient ($dL/d\theta$) of the SRM. However, the Cubic logic ensures these dips remain rounded, which is critical for minimizing acoustic noise.
- **High-Frequency Ripple:** The ”fuzz” observed on the current and torque signals is the expected result of the hysteresis (bang-bang) control. The regulator continuously switches the inverter states to maintain the current within the defined 0.5 A tolerance band.

## 4.5 Next Milestone

The current milestone successfully validates the fundamental operation of the 6/4 SRM using the Cubic TSF control strategy. The subsequent phase of this research will focus on a comparative assessment of alternative Torque Sharing Function configurations, specifically the Linear (TSF 1) and Exponential profiles. This analysis is intended to provide a quantitative measure of the torque ripple reduction achievable through different distribution logics.

Furthermore, the study will investigate the integration of advanced outer-loop speed controllers to enhance disturbance rejection and transient performance. The project aims to evaluate the effectiveness of various control topologies, those may include:

- Sliding Mode Control (SMC): To provide superior robustness against parameter uncertainties and external disturbances.
- Fuzzy Logic Control (FLC): To effectively manage the non-linear torque-speed characteristics of the SRM without a precise mathematical model.
- Model Predictive Control (MPC): To optimize the speed trajectory while respecting the current and voltage constraints of the power converter.

These reconfigurations will be analyzed to determine the optimal control architecture for high-precision applications.

# Bibliography

[1] MathWorks, “Srm commutation and operation,” 2024. Accessed: April 2026.